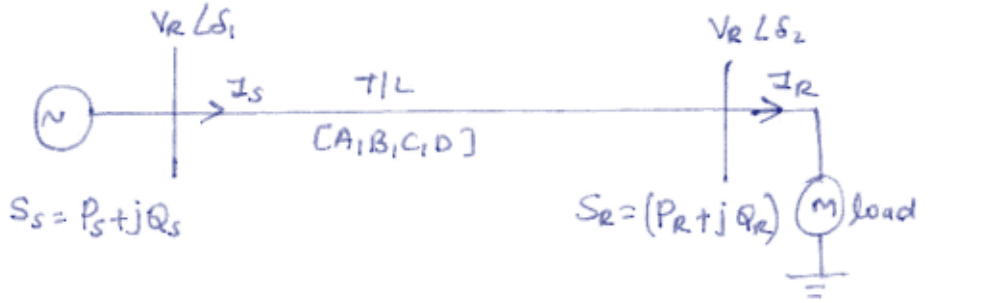
Power Flow Through Transmission Line

Power flow through the T/L :



Generator - [Vs at delta1, Is] - T/L [A, B, C, D] - [VR at delta2, IR] - load; Ss = Ps + jQs, SR = PR + jQR

S_s = Sending end apparent power ; P_s = Active power ; Q_s = Reactive power

Q_R = Receiving end Reactive power ; P_R = Receiving end Active power ; S_R = Receiving end Apparent power.

δ_1, δ_2 = load (or) power angle ; $\delta = (\delta_1 - \delta_2)$

Assume :

$$A = |A| \angle \alpha \ , \ D = |D| \angle \alpha \ [\because A = D] \ , \ B = |B| \angle \beta \ , \ C = |C| \angle \Delta \ , \ AD - BC = 1$$

Receiving end power :

$$P_R = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) - \left|\frac{A V_R^2}{B}\right| \cos(\beta - \alpha) \quad - (1)$$

$$Q_R = \left|\frac{V_S V_R}{B}\right| \sin(\beta - \delta) - \left|\frac{A V_R^2}{B}\right| \sin(\beta - \alpha) \quad - (2)$$

Trick : cos -> sin ; $\delta = \delta_1 - \delta_2$

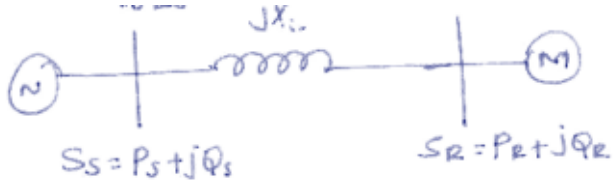
Sending end power :

$$P_S = \left|\frac{D V_S^2}{B}\right| \cos(\beta - \alpha) - \left|\frac{V_S V_R}{B}\right| \cos(\beta + \delta) \quad - (3)$$

$$Q_S = \left|\frac{DV_S^2}{B}\right| \sin(\beta-\alpha) - \left|\frac{V_S V_R}{B}\right| \sin(\beta+\delta) \quad - (4)$$

(1), (2), (3), (4) are applicable for all types of lines (short, med, long)

Short T/L with only X_L



Generator Vs at delta, jXL, VR at 0 degrees, motor; Ss = Ps + jQs, SR = PR + jQR

$$\begin{bmatrix} A\angle\alpha & B\angle\beta \\ C\angle\Delta & D\angle\alpha \end{bmatrix} = \begin{bmatrix} 1\angle 0^\circ & X_L\angle 90^\circ \\ 0 & 1\angle 0^\circ \end{bmatrix} \quad - (5)$$

Substitute (5) in (1), (2), (3), (4)

$$P_R = P_S = \left|\frac{V_S V_R}{X_L}\right| \sin\delta, \quad Q_S = \left|\frac{V_S^2}{X_L}\right| - \left|\frac{V_S V_R}{X_L}\right| \cos\delta$$

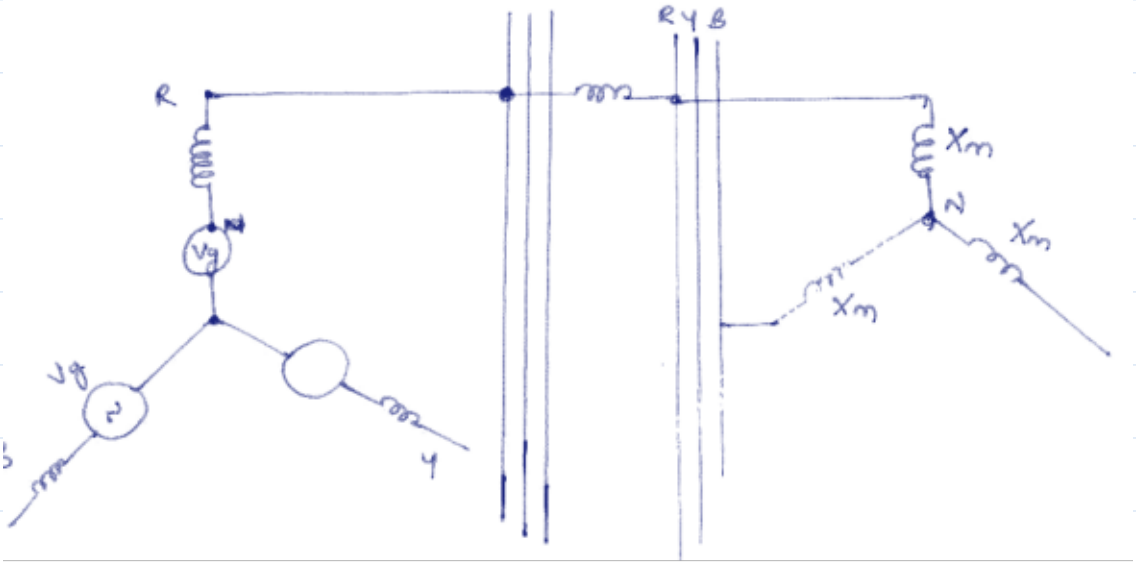
$$Q_R = \left|\frac{V_S V_R}{X_L}\right| \cos\delta - \left|\frac{V_R^2}{X_L}\right|$$

3-ϕ Gen, motors, T/F => MVA capacity => 3-ϕ power

3-ϕ T/L -> $R_1, L_1, C_1, G_1, A, B, C, D$ } per phase quantities

3-ϕ T/L => Vtg => line vtg (rms)

V_{SL} = 132, 220, 400 kV



Three-phase generator (Vg, star) - R, Y, B line with reactance - star-connected motor (Xm)

$$P_{Rph} = \frac{V_{Sph} \cdot V_{Rph}}{X_{Lph}} \sin(\delta)$$

$$P_{Rph} = \frac{\frac{V_{SL}}{\sqrt{3}} \cdot \frac{V_{RL}}{\sqrt{3}}}{X_{Lph}} \sin(\delta)$$

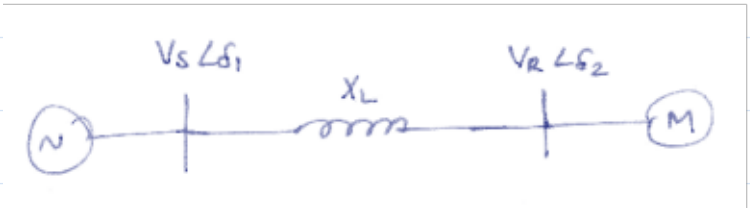
$$\boxed{P_{R3\phi} = 3P_{Rph} = \left/ \frac{V_{SL} V_{RL}}{X_{Lph}} \right/ \sin(\delta)}$$

Objective of Transmission line :

(1) Power Transfer capability must be higher with minimum line losses and maximum η (efficiency).

(2) Vtg Regulation is zero or minimum.

Analysis of objective (1) :



Generator Vs at delta1 - XL - VR at delta2 - motor

$$P_S = P_R = \left|\frac{V_S V_R}{X_L}\right| \sin\delta, \quad \text{Assume } |V_S| = |V_R|$$

$$\uparrow P_S = \frac{\uparrow |V_S^2|}{|X_L| \downarrow} \sin\delta \uparrow$$

Advantage of higher Transmission line.

(1) Power transfer capacity of line increased : $P_S \propto V_S^2$

(2) Tr. line losses are reduced : $P = VI\cos\phi$, $I = \frac{P}{V\cos\phi}$, P_L = Tr. line losses = I^2R

$$\boxed{P_L = \frac{P^2 R}{V^2 \cos\phi}}$$

$$\boxed{P_L \propto \frac{1}{V^2}}$$

condition: P, R, $\cos\phi$ = const.

(3) Area (or) Volume of conductor is reduced

$$R = \frac{\rho l}{A} \Rightarrow A = \frac{\rho l}{R}$$

$$\boxed{A = \frac{P^2 \rho l}{V^2 \cos^2\phi \cdot P_L}}$$

$A \propto \frac{1}{V^2}$, Volume $K = Al$, $K \propto A \propto \frac{1}{V^2}$

condition: P, P_L, ρ, l, $\cos\phi$ = const.

NOTE : Because of above advantages Tr. vtg increased from 33 - 132 - 220 - 400 - 765 kV

Methods of Reducing X_L :

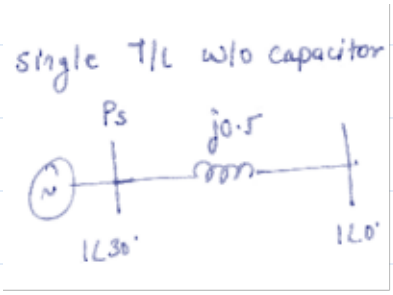
$$\uparrow P_s \propto \frac{1}{X_L \downarrow}$$

(1) Bundled conductor

(2) parallel Tr. line

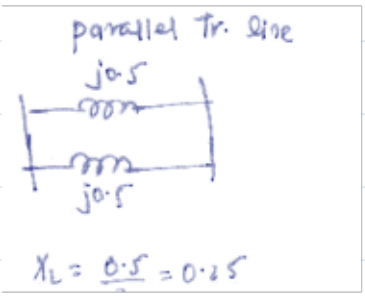
(3) Series Capacitor

Single T/L w/o capacitor : $P_{S1} = \left|\frac{1 \times 1}{0.5}\right| \sin(30^\circ) = 1 \text{ pu} = P_1$



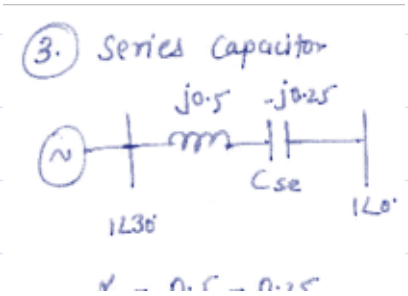
Single line: jXL = j0.5, Vs = 1 at 30 degrees, VR = 1 at 0 degrees

Parallel Tr. line : $X_L = \frac{0.5}{2} = 0.25$; $P_S = \left|\frac{1 \times 1}{0.25}\right| \sin(30^\circ) = 2 \text{ pu} = 2P_1$



Two parallel lines, each j0.5

Series capacitor : $X_L = 0.5 - 0.25 = 0.25$; $P_S = \left|\frac{1 \times 1}{0.25}\right| \sin(30^\circ) = 2 \text{ pu} = 2P_1$

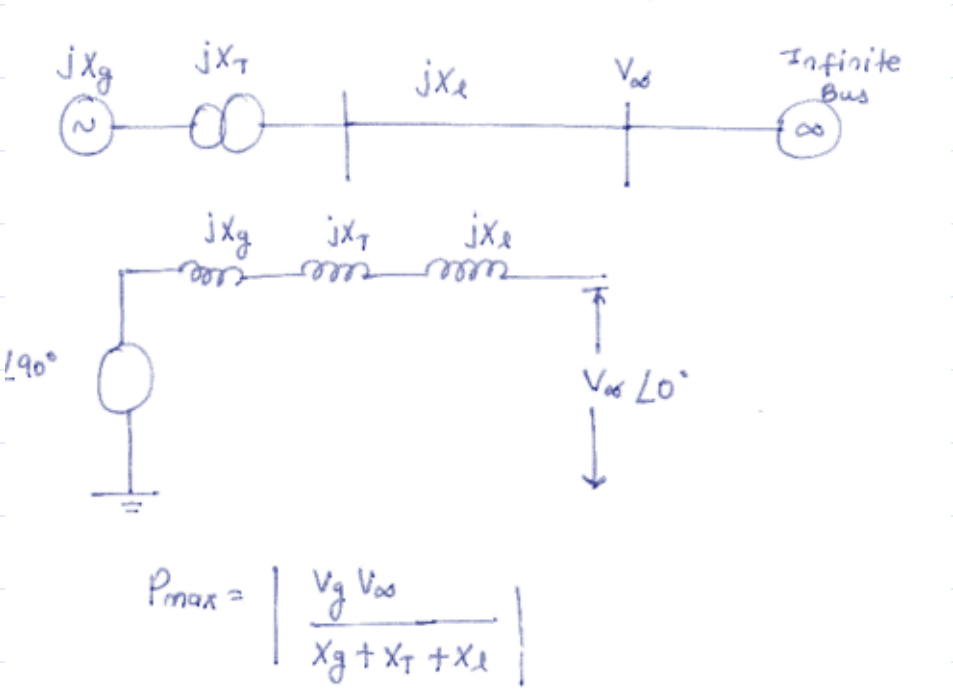


j0.5 in series with -j0.25 (Cse)

DAY-4 [12/9/25]

Calculation of P_{max} :

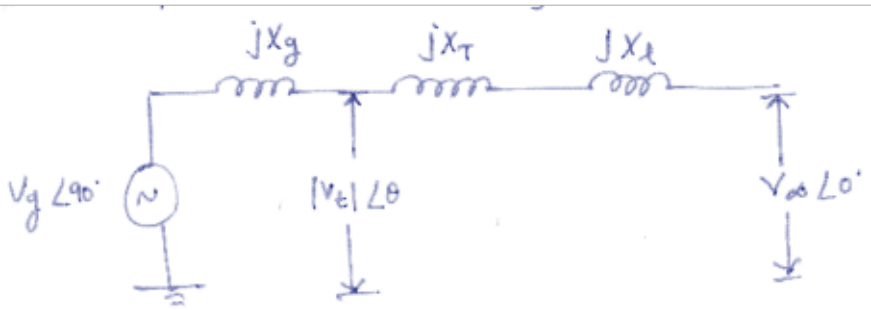
Case 1 : Generator internal (or) No load vtg or vtg behind sub-transient reactance is given



Generator jXg - transformer jXT - line jXl - infinite bus V_inf; equivalent: Vg at 90 degrees, jXg, jXT, jXl, V_inf at 0 degrees

$$P_{max} = \left| \frac{V_g V_\infty}{X_g + X_T + X_l} \right|$$

Case 2 : Generator Terminal vtg is given :



Vg at 90 degrees, jXg, terminal |Vt| at angle theta, jXT, jXl, V_inf at 0 degrees

$$I = \frac{V_g \angle 90^\circ - V_t \angle \theta}{jX_g} = \frac{V_t \angle \theta - V_\infty \angle 0^\circ}{j(X_T + X_l)}$$

Separate Real and Imag part and calculate

$$\cos\theta = \left| \frac{V_\infty X_g}{V_t (X_g + X_T + X_l)} \right|$$

Steps for calculation of P_{max} :

Calculate :

(1)

$$\theta = \cos^{-1} \frac{V_\infty X_g}{V_t (X_g + X_T + X_l)}$$

(2)

$$I = \frac{V_t \angle\theta - V_\infty \angle 0^\circ}{j(X_T + X_l)}$$

(3)

$$V_g = V_t \angle\theta + I(jX_g) = |V_g| \angle 90^\circ \rightarrow \text{cross check angle}$$

(4)

$$P_{max} = \left| \frac{V_g V_\infty}{(X_g + X_T + X_l)} \right|$$

Eg: Find P_{max} if

(a) $V_g = 1.2$ p.u , $V_\infty = 1$ pu , $X_g = 0.1$, $X_T = 0.15$, $X_l = 0.25$

(b) $V_t = 1.2$ p.u , $V_\infty = 1$ p.u , $X_g = 0.1$, $X_T = 0.15$, $X_l = 0.25$

(a)

$$P_{max} = \frac{1.2 \times 1}{0.1 + 0.15 + 0.25} = 2.4$$

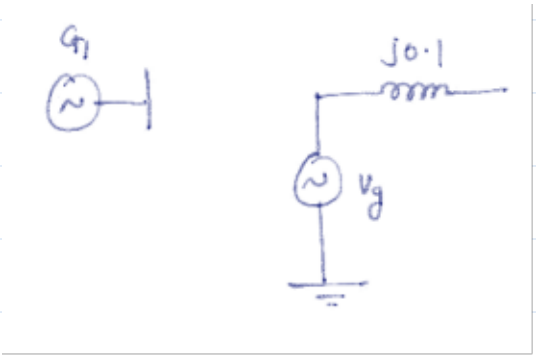
(b)

$$\theta = \cos^{-1} \left[\frac{1 \times 0.1}{1.2(0.5)} \right] = 80.4^\circ$$

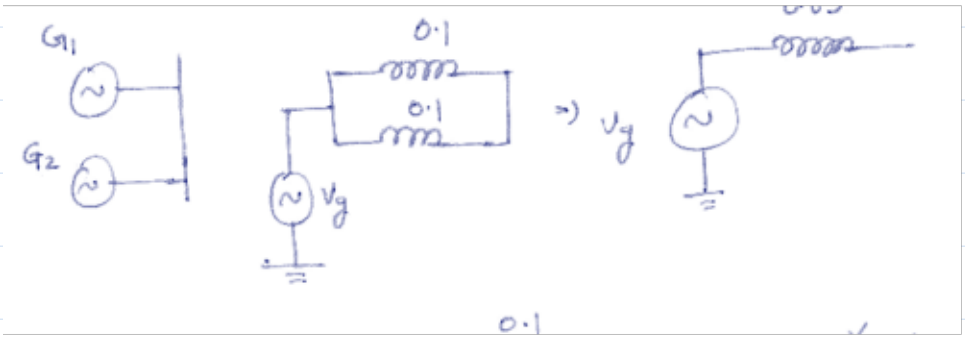
$$I = \frac{1.2 \angle 80.4 - 1 \angle 0^\circ}{j(0.15 + 0.25)} = 3.57 \angle 34.06^\circ$$

$$V_g = 1.2\angle 80.4^\circ + j(0.1)(3.57\angle 34.06^\circ) = 1.479\angle 90^\circ$$

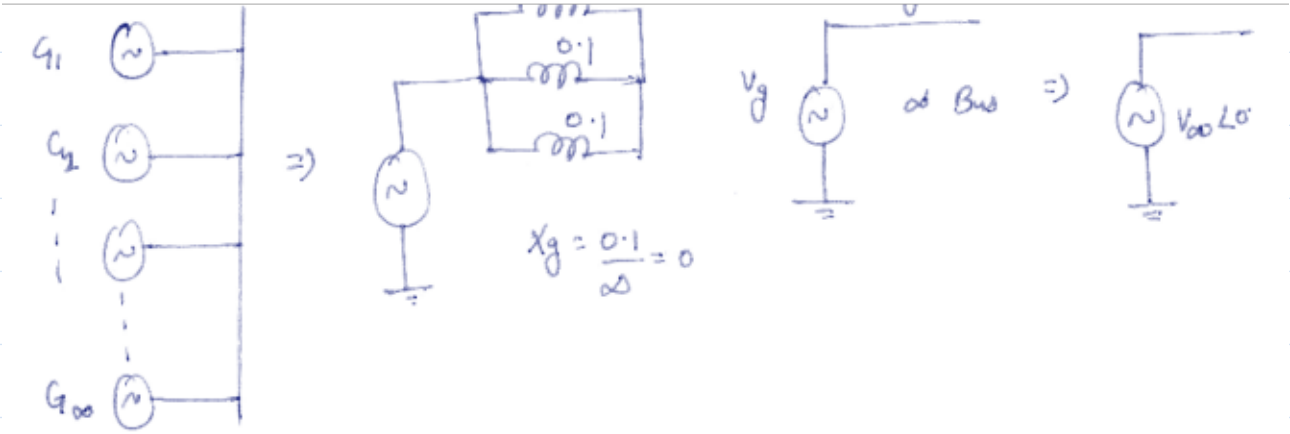
$$P_{max} = \frac{1.479 \times 1}{0.5} = 2.95 \text{ p.u}$$



One generator G1: Xg = j0.1 behind Vg



Two generators G1, G2 in parallel: 0.1 // 0.1 = 0.05



Many generators G1 ... G_inf in parallel: Xg = 0.1/inf = 0, so the bus becomes an infinite bus V_inf at 0 degrees

* Infinite bus has const. vtg, const. frequency; Internal Reactance of infinite bus is assumed as zero.

Importance of load (or) power Angle (δ)

For already constructed Tr. line, V_s, V_R and X_L are constant. For continuously varying active power demand P_R, load angle (δ) is selected as a variable by varying the governing valve of the turbine. Practical value of (δ) is maintained between 30 to 55° due to stability limitations.

$$\uparrow P_R = \left| \frac{V_S V_R}{X_L} \right| \sin\delta \uparrow = P_{max} \sin\delta$$

δ -> Angle between Stator and Rotor magnetic field.

* For protection of generator from the loss of synchronism, pole slip or (out of step) relay are used.